Transformer u(t) Reproduction Report

Fixed-initial-condition time-dependent control under PMP/KKT optimality conditions

This report focuses on the manuscript's time-dependent control strategy u_theta(t): [0,T] -> [0,u_max] for the fixed initial condition. The state-dependent feedback extension u(t,N) is not included here.

1. Model and Training Objective

We use the population dynamics and cost from the manuscript:

$$\dot{N}_i(t) = \big(r_i - \phi_i u(t) - M_i G(N(t))\big) N_i(t) \tag{1}$$

$$G(N) = \log\left(1 + \frac{1}{m}\sum_{k=1}^{m} N_k\right) \tag{2}$$

$$J(u) = \alpha^\top N(T) + \int_0^T \big(\beta^\top N(t) + \gamma u(t)\big)\, dt \tag{3}$$

Reported parameters: T=10, m=21, u_max=3, alpha=1, beta=0.1, gamma=20, and N_i(0)=10. The control is represented by a small Transformer encoder over normalized time.

Given u_theta(t), we roll out N_theta(t), solve the costate equation backward, and compute the switching function

$$\psi(t) = H_u(N, \lambda, u) = \gamma - \sum_i \phi_i \lambda_i(t) N_i(t) \tag{4}$$

component	role
non-singular KKT loss	enforces u=0 when psi>0 and u=u_max when psi<0
singular loss	near psi=0, matches u_theta(t) to the singular candidate u_sing(N(t))

Here q(t) is a smooth weight that switches between the singular condition near psi(t)=0 and the boundary KKT condition away from psi(t)=0.

2. Learned u(t), N(t), and N(u)

The learned control starts high, transitions to a lower interior/singular-like region, and increases again near the terminal portion.

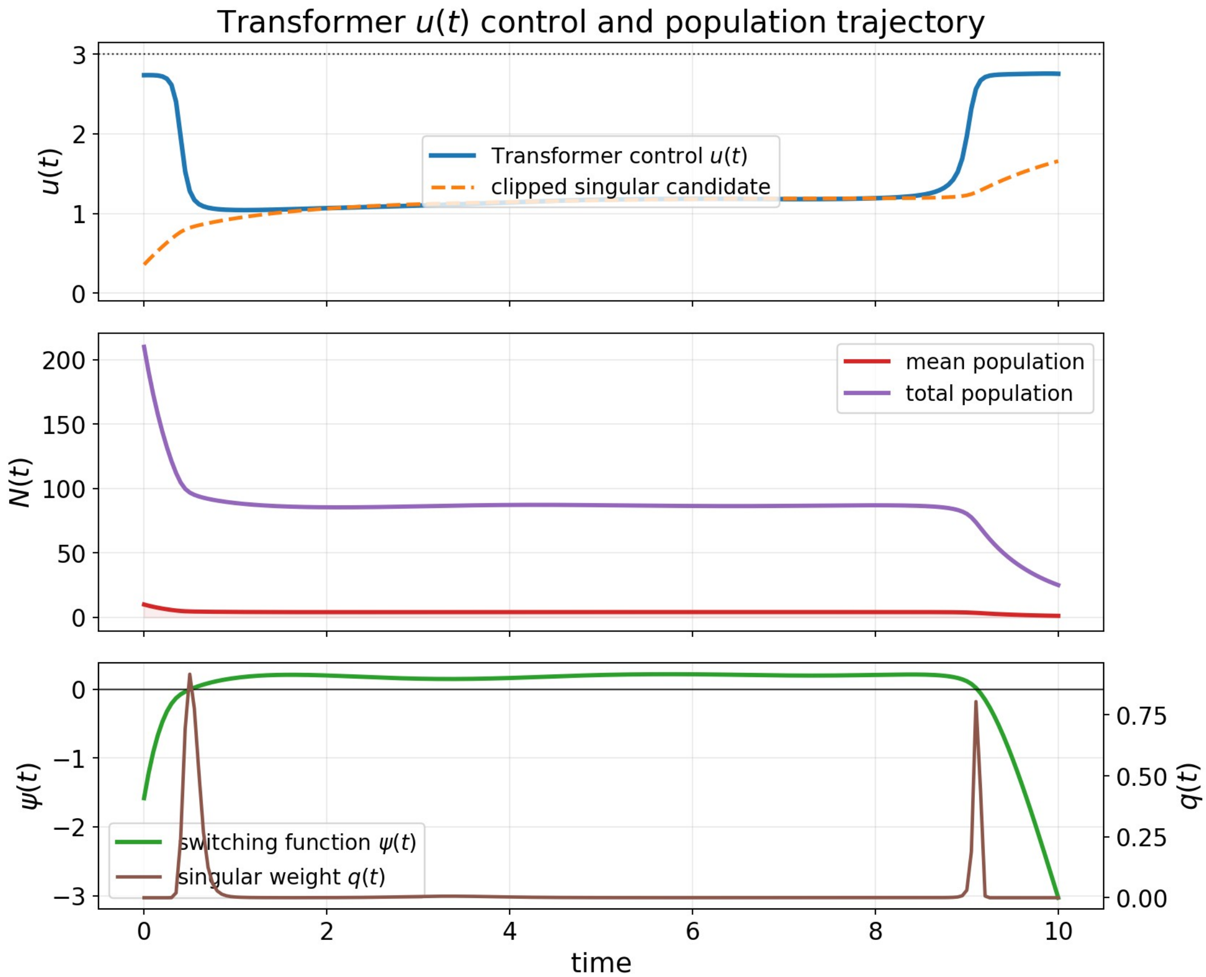

Figure 1. Learned Transformer control u(t), population trajectory N(t), switching function psi(t), and singular weight q(t).

The N(u) phase plot below uses the total population across all 21 subpopulations.

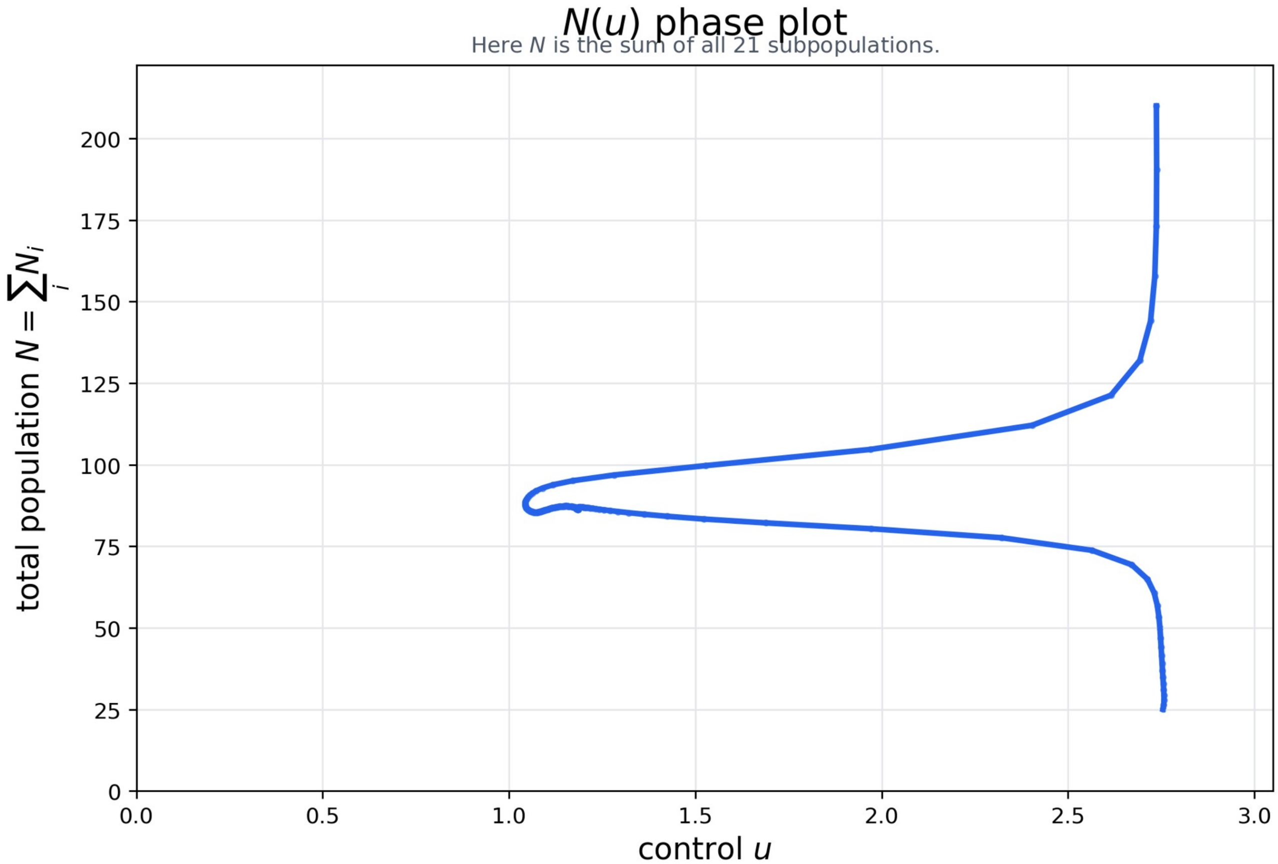

Figure 2. N(u) phase plot using the total population across the 21 subpopulations.

3. Training Loss Trajectories for PMP/KKT Conditions

The table reports the smallest recorded training loss. In this experiment, the training loss is the manuscript's PMP/KKT optimality gap, composed of the singular condition and the non-singular Hamiltonian minimization condition.

metric	value
total training loss / PMP-KKT optimality gap	0.02637
singular-condition training loss	0.00167
non-singular Hamiltonian-minimization training loss	0.02471
objective J during training	384.76
range and mean of u_theta(t)	min 1.038, max 2.758, mean 1.372
terminal mean state	1.207

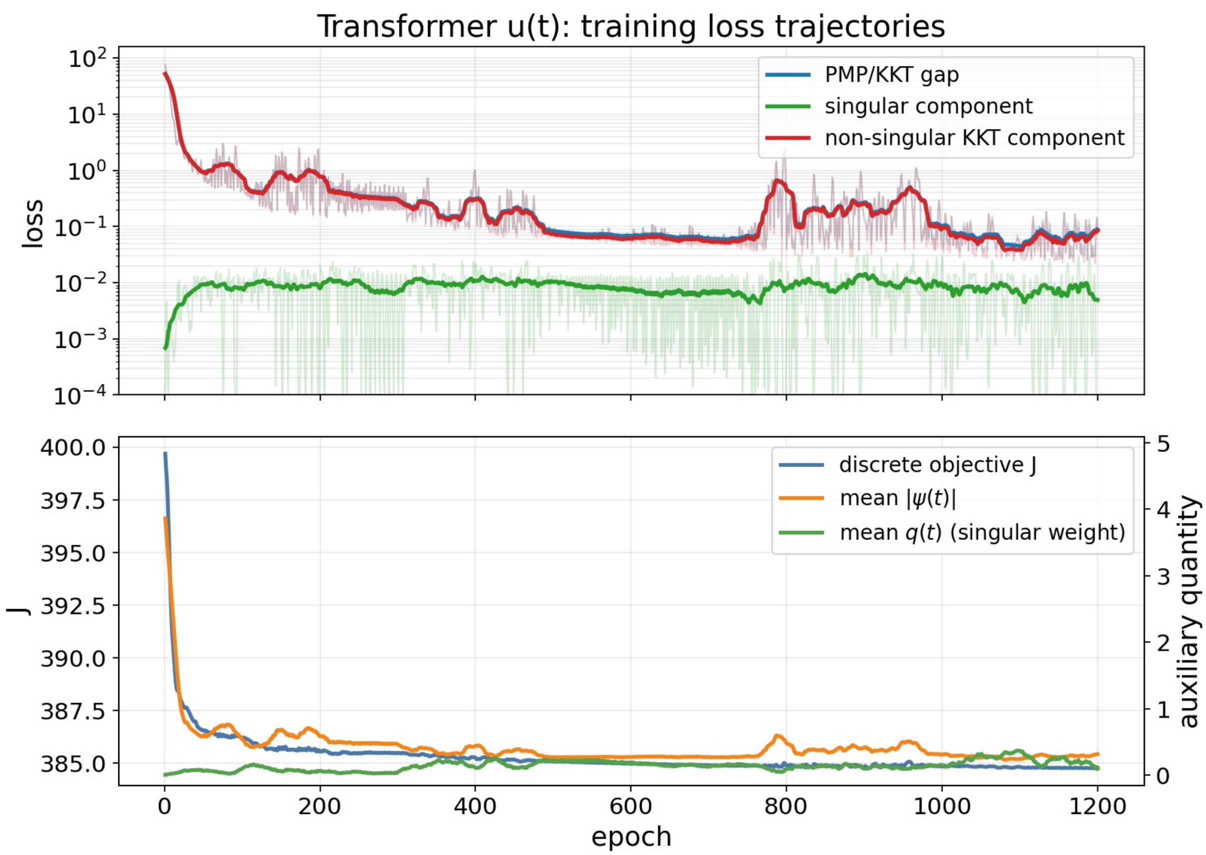

Figure 3. Training trajectories of the total PMP/KKT optimality gap and its singular and non-singular components.

The final pointwise condition plot below uses a smooth weight q(t) to separate the two regimes: near psi(t)=0 it emphasizes the singular-condition error; away from psi(t)=0 it emphasizes the boundary KKT error.

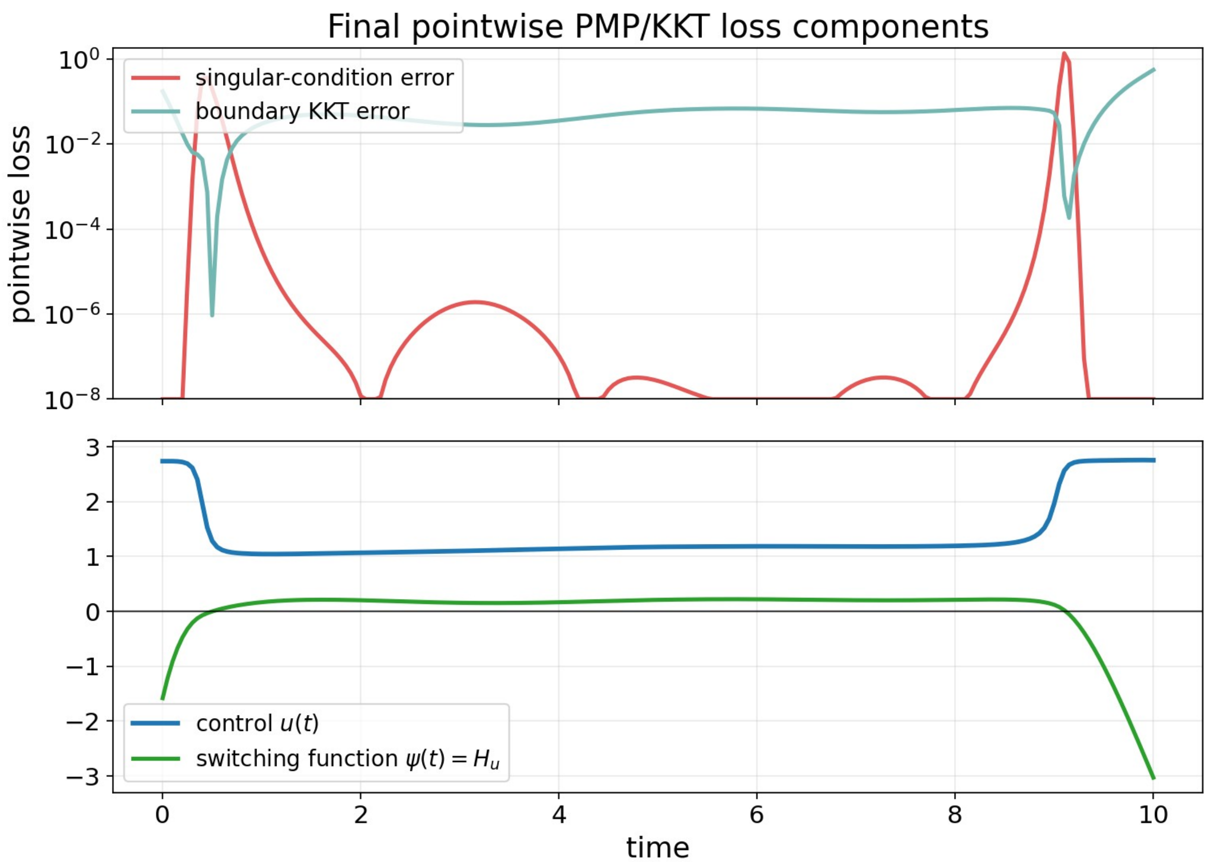

Figure 4. Pointwise singular-condition error and boundary KKT error along the final learned trajectory.

4. Comparison with Related Work

For the fixed-initial-condition u(t) reproduction, all numerical comparisons keep the same initial condition and compare time-dependent controls u(t). Several related-work papers in the manuscript target value-function or feedback formulations; those are important related formulations, but they are not the same numerical task as this u(t) reproduction.

method family	learned object / formulation	relation to this report
HJB / BSDE methods [1,2,7]	value function V(t,N) or HJB PDE solution	state-domain feedback/value formulation; not a direct numerical comparison for fixed u(t)
Neural-PMP [3]	control sequence via forward rollout, backward costate recursion, and Hamiltonian-gradient updates	closest related-work method for the present u(t) experiment; implemented numerically below
DeepONet / PINN policy iteration [5,6]	policy evaluation and improvement for HJB-type equations	related feedback/value-learning formulation; separate from this fixed-trajectory u(t) layer
classical chemotherapy OC [4]	PMP and singular-control structure	source of the singular-control condition used in the manuscript

Among these, [3] is the closest numerical comparison. Here [3] refers to Gu, Xiong, and Chen, Pontryagin Optimal Control via Neural Networks (arXiv:2212.14566). Their Neural-PMP method first learns a differentiable dynamics model and then updates a control sequence using PMP gradients. Since the dynamics are already known in the manuscript setting, we compare with the corresponding known-dynamics u(t) update: forward state integration, backward costate recursion, and Hamiltonian-gradient updates of a discrete control sequence.

method	optimization condition	objective J	J - direct J	PMP/KKT gap	role
direct minimization of J	minimize discretized J on a time mesh	386.438	0.000	1.016	cost comparison
Transformer u(t), 6 trainings	manuscript PMP/KKT optimality-gap loss	386.738 +/- 0.045	0.300 +/- 0.045	0.463 +/- 0.148	reproduction
best Transformer training	same as above	386.695	0.257	0.207	best Transformer result
Neural-PMP [3]	PMP-gradient update for u(t) with known dynamics	386.986	0.548	8.574	related-work comparison
constant u=1.5	fixed control	400.403	13.965	76.556	simple control
provided repository output	provided s.csv	422.670	36.232	433.349	provided output file

* For comparison, after u(t) is fixed, N(t) and J are recomputed on a common fine time mesh using fourth-order Runge-Kutta. This is only the numerical evaluation of the manuscript objective.

The Transformer trainings are stable across independent initializations and remain close to direct minimization of J. Compared with Neural-PMP [3], the Transformer has both lower objective J and a smaller PMP/KKT optimality gap in this fixed-initial-condition u(t) experiment.

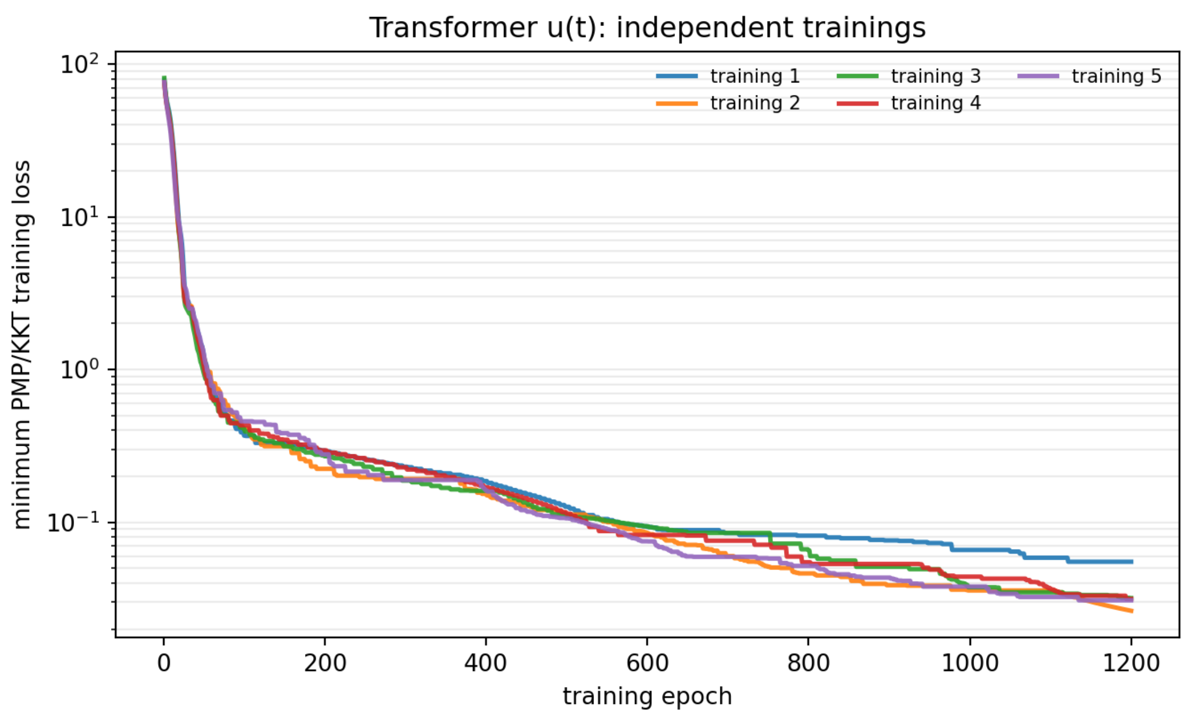

Figure 5. Convergence of the Transformer u(t) PMP/KKT training loss across independent trainings.

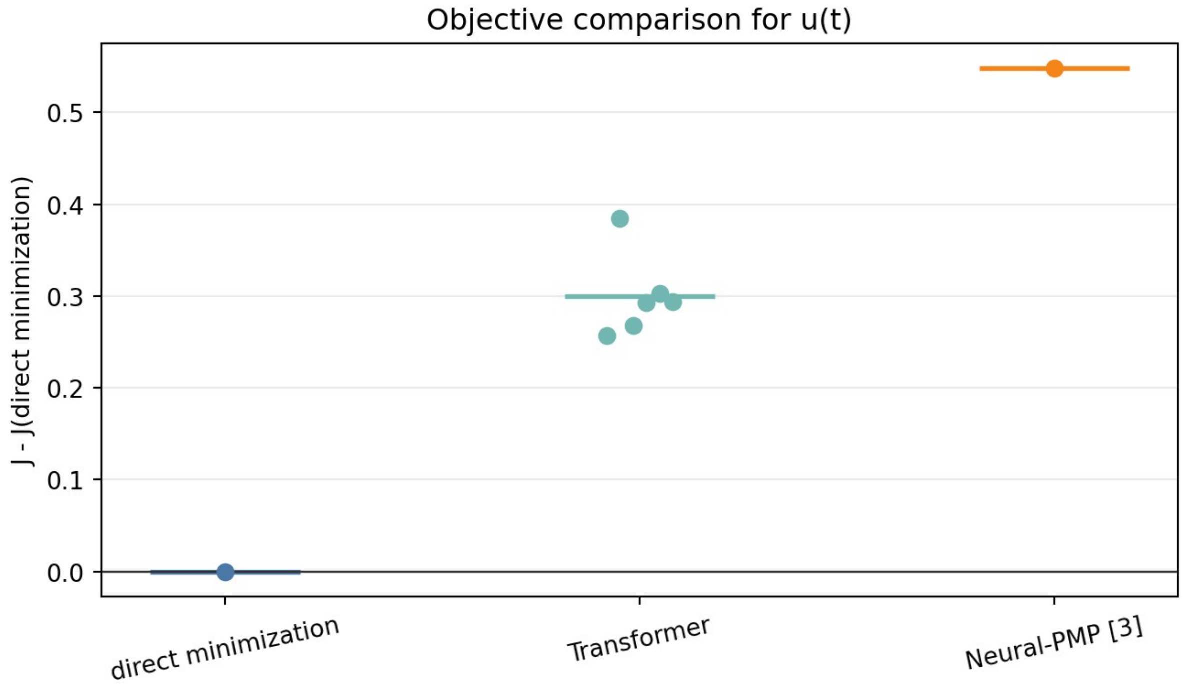

Figure 6. Difference in J from direct minimization for the main u(t) comparison.

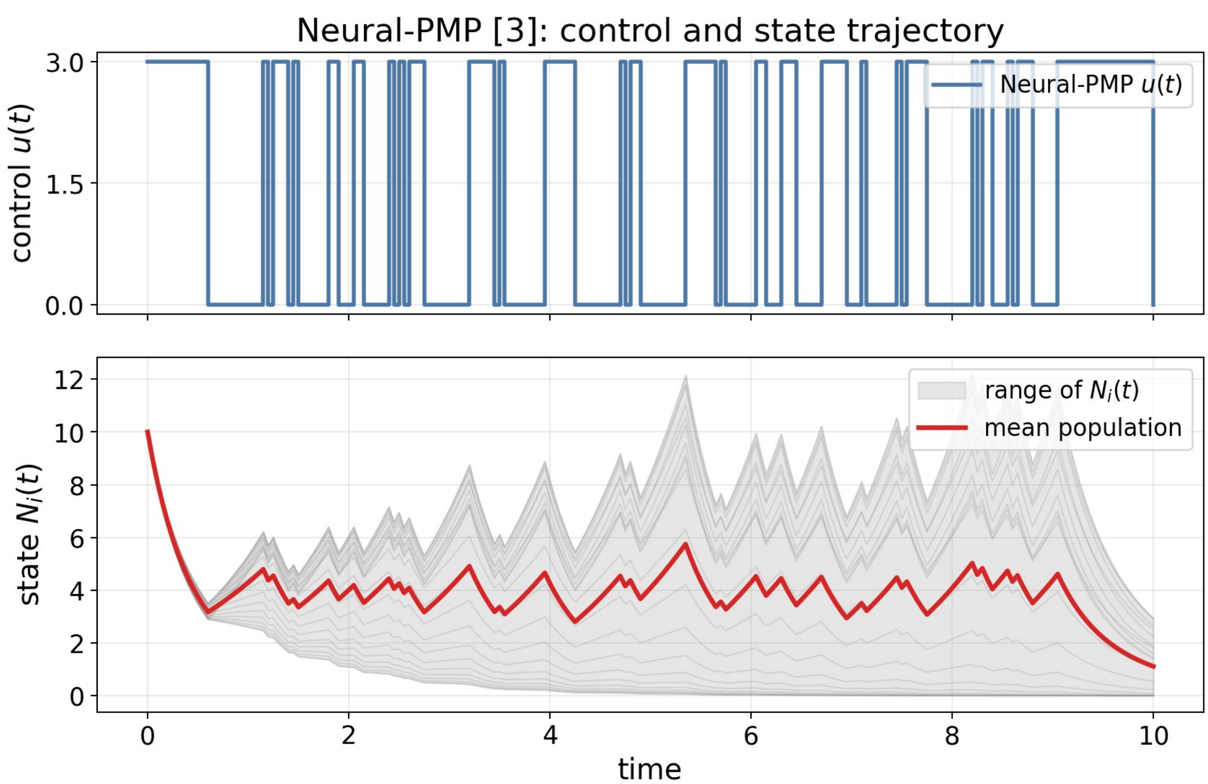

Figure 7. Neural-PMP [3]: control sequence and resulting state trajectory.

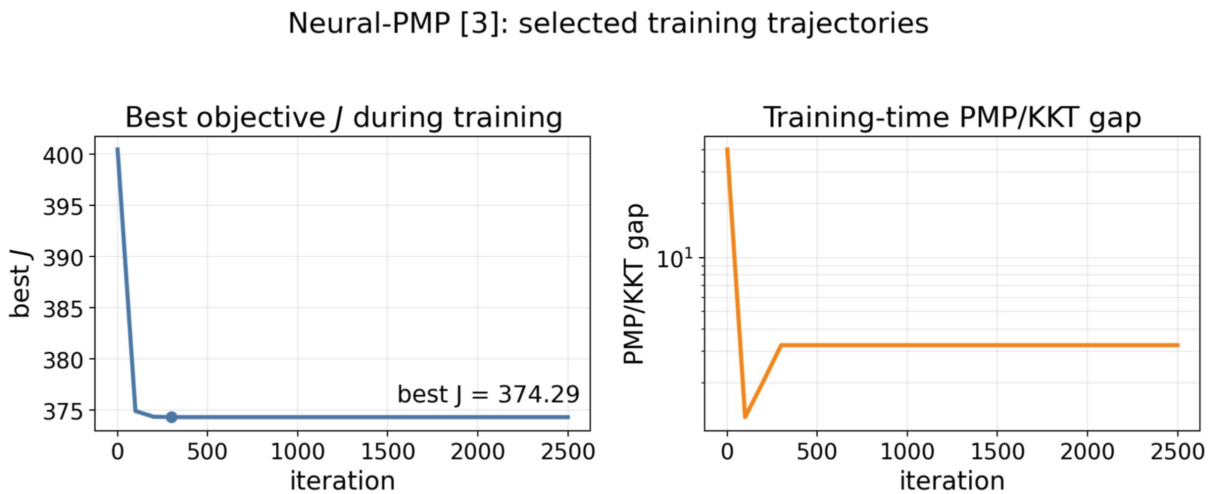

Figure 8. Neural-PMP [3]: selected training trajectories.

5. Parameter Sensitivity

We also tested whether the u(t) reproduction is tied to the nominal parameter choice. In each row below, one parameter is changed while the others remain at the reported setting. The Transformer result is averaged over three independent trainings; Neural-PMP [3] is the best result from the same search protocol.

condition	direct J	Transformer J	Transformer gap	Neural-PMP gap
nominal	386.706	386.826 +/- 0.052	0.031%	0.240%
beta=0.05	325.457	325.713 +/- 0.030	0.079%	0.169%
beta=0.20	444.860	444.923 +/- 0.006	0.014%	0.160%
gamma=10	230.385	230.485 +/- 0.011	0.044%	0.168%
gamma=40	614.915	615.043 +/- 0.021	0.021%	0.042%
alpha=0.5	370.513	370.482 +/- 0.023	-0.008%	0.135%
alpha=2	404.406	404.600 +/- 0.033	0.048%	0.247%
N0=5	371.118	371.168 +/- 0.037	0.013%	0.220%
N0=20	404.839	405.201 +/- 0.074	0.089%	0.174%

* In this sensitivity table, direct J is obtained by direct minimization on an n=400 time mesh and then recomputed on the common fine time mesh. The small negative alpha=0.5 gap is within this numerical tolerance and should be read as a tie, not as beating the optimum.

Across these one-factor tests, the Transformer remains within about 0.09% of direct minimization. Neural-PMP [3] is also close in objective value, but the Transformer is consistently closer to direct minimization and has a smaller PMP/KKT optimality gap.

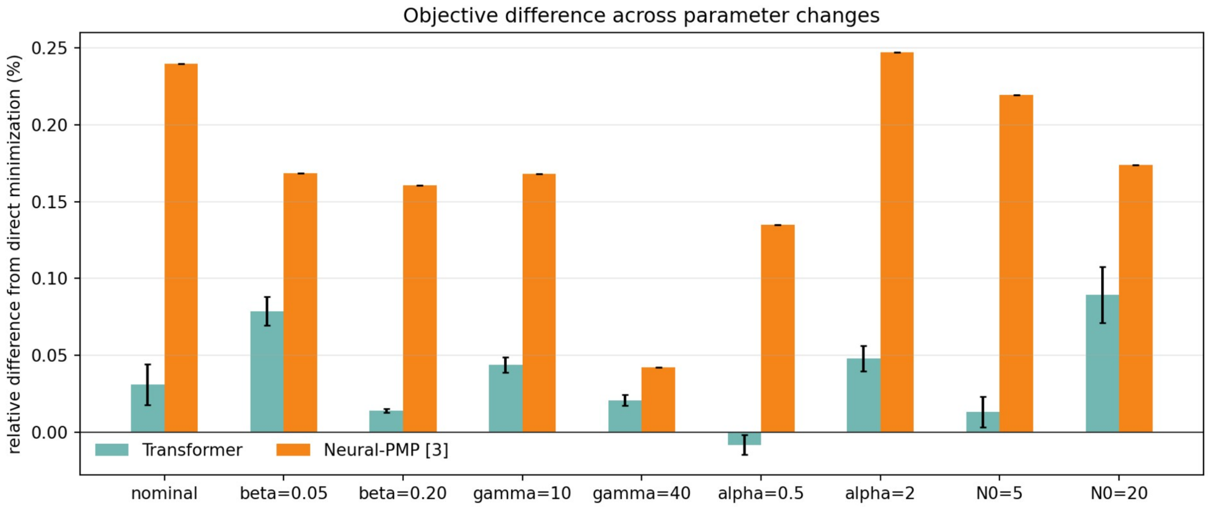

Figure 9. Difference in objective J from direct minimization across parameter and initial-condition changes.

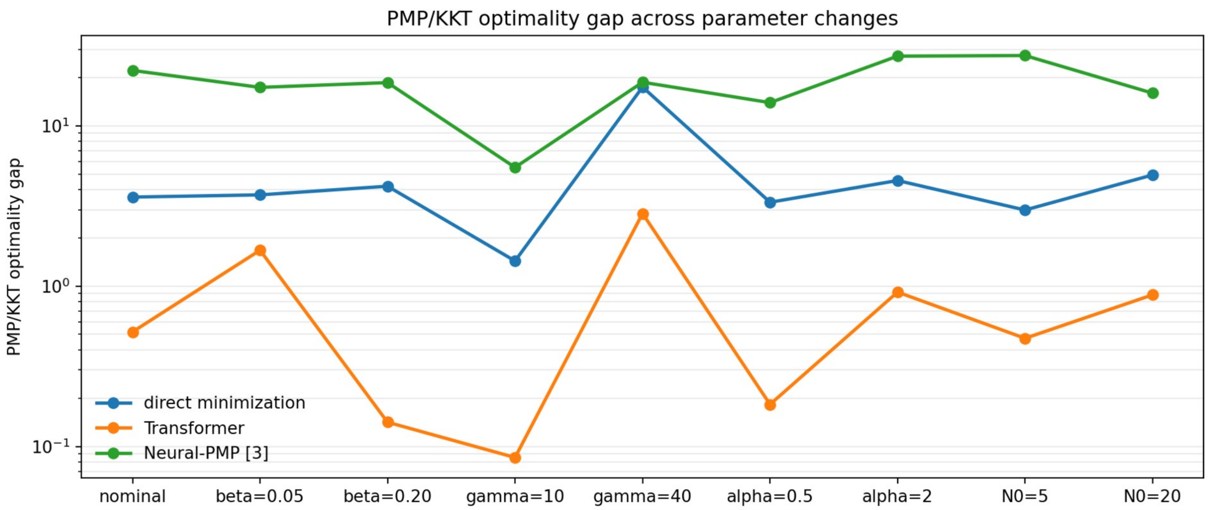

Figure 10. PMP/KKT optimality gap across the same sensitivity conditions.

6. Conclusion

The requested u(t) reproduction is complete. The Transformer control trained with the manuscript's PMP/KKT optimality-gap loss is smooth, satisfies the control bounds, and reduces the training optimality gap from about 76 to 0.026 in the best training. Across six Transformer trainings in the nominal setting, the recomputed objective is 386.738 +/- 0.045, close to direct minimization of J at 386.438 and better than Neural-PMP [3] in this comparison. The parameter sensitivity experiments further show that the same u(t) training procedure remains close to direct minimization under changes in beta, gamma, alpha, and N0. The state-dependent extension u(t,N) is left for separate work because it requires different optimality conditions.